

Connected Vehicles Travel Time Prediction in a Scenario with Adaptive Traffic Light Control

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Abstract— The paper is devoted to the short-term travel time prediction of individual connected vehicles in a regulated road network with adaptive control of traffic lights. The estimation of the total travel time combines both the travel time along road network links, obtained by a neural network model, and the waiting time at regulated intersections. At the first stage, it is proposed to use the model based on a neural network to estimate the travel time along the road links of the transportation network. At the second stage, the phase of the traffic light is predicted using the adaptive control method. Finally, the waiting time at the intersection is calculated based on the predicted arrival time of the vehicle at the intersection and the duration of the traffic light phase. Experimental results in a microscopic simulation environment allow us to conclude that the proposed approach outperforms baseline methods in terms of the mean absolute error criterion.

Keywords—connected vehicles, travel time prediction, adaptive traffic light control, machine learning

I. INTRODUCTION

The active development of intelligent transportation systems requires the solution of a large number of research problems related to data analysis [1], [2]. One of the main tasks in the area of transport systems is the short-term prediction of the average characteristics of the traffic flow, such as speed and flow density. Predictive traffic flow distribution is a key factor in the development of effective control strategies in intelligent transport systems, incl. for vehicle navigation, traffic light adaptive control, and reduction of the traffic congestion level in general. Accordingly, the relevance of the problem entailed the development of a large number of methods and algorithms for solving it, from simple linear regression models to deep neural networks that allow taking into account complex spatiotemporal dependencies in the data [3], [4].

At the same time, due to the active development of connected and autonomous vehicles, the problem of predicting the movement of individual vehicles is relevant. However, it is necessary to take into account not only the travel time of movement along the links of the transport network but also the waiting time at intersections regulated by traffic lights. According to [5], the delay at the intersection can be up to 40% of the total travel time. Moreover, the travel time of even two consecutive vehicles on the same road network link can differ significantly if they reach the intersection at different phases of the traffic light cycle. For instance, waiting at the red phase of a traffic light increases the vehicle's travel time. Consequentially, it is essential to predict the arrival time of individual vehicles at the intersection and estimate the waiting time on it, taking into account the predicted traffic light phase.

The article considers the problem of predicting the travel time of individual connected vehicles (CV), for which the positions on the road network links at each moment of time, the speed, and the route are considered known. To estimate the total vehicle travel time, the proposed approach combines the travel time along road network links with the waiting time at regulated intersections. In addition, a transport network with adaptive traffic light control is considered, the purpose of which is to select the traffic light phase based on the observed traffic situation.

II. LITERATURE REVIEW

To review the papers devoted to the problem considered in this article, we consider separately the travel time prediction problem and the problem of adaptive traffic light control.

A. Travel Time Prediction

An overview of the main solutions to the traffic prediction problem along with existing unsolved problems was presented in [3]. Modern works devoted to the short-term prediction of traffic flows are mainly based on deep learning models. These models include convolutional neural networks, recurrent neural networks, LSTM networks, etc. An overview of deep learning-based approaches, a performance analysis of different methods in various real-world scenarios, the main factors affecting the results of the models, and open challenges was presented in [4], [6]. In [7], the authors provided a systematic review of methods and algorithms for speed prediction on three levels: traffic speed prediction on a macro level, lane speed prediction on a mesoscopic level, and vehicle speed prediction on a microscopic level. In addition, the authors summarize evaluation metrics, datasets, and open-source implementation of the methods under consideration.

In [8], the authors considered a mixed traffic flow. The speeds of connected and autonomous vehicles were used as additional parameters in the traffic flow prediction model. In [9], it was proposed to use a two-level prediction approach. At the first step, the average traffic speed is predicted by using a neural network model. The speed prediction for an individual vehicle is performed by an algorithm based on Hidden Markov models, taking into account the estimated traffic speed. In [10], the authors predicted the movement of individual vehicles based on a kinematic wave model. The travel time of each vehicle is the sum of the free travel time along the road segment and the delay time at the intersection. However, a fixed cycle is used as the traffic light mode, not adaptive control.

The problem of short-term arrival time prediction of public transport is similar in formulation to the one under consideration [11]. In [12], the authors provided a systematic survey on the arrival time prediction problem, described open problems, and future research directions. The arrival time

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prediction of a bus at stops using the LSTM model was studied in [11], [13]. In [14], the authors proposed a model for predicting the arrival time at a stop, taking into account both the observed traffic flow and delays at intersections.

B. Adaptive Traffic Light Control

Since the travel time prediction problem in this paper is considered in a scenario with adaptive traffic light control, in this subsection we provide an overview of modern approaches to traffic light control.

In [15], the authors provided a review of computational intelligence methods for optimizing traffic signal timing, including simulation-based approaches. A survey on methods for traffic signal control, open issues, and challenges was presented in [16].

Recently, much attention has been drawn to the use of reinforcement learning (RL) methods in various research and practical problems, including the problem of traffic light control. In [17], the authors provided a comprehensive review of RL methods applied to solve the traffic signal control problem in urban networks. A comparison of Deep RL methods and architectures as well as guidelines for its application was presented in [18].

Coordinated traffic signal control based on multi-agent RL was proposed in [19]. In [20], the authors proposed a computationally effective deterministic approach based on maximizing the predicted number of vehicles passing through an intersection during a selected traffic light phase. This approach was modified in [21] to take into account waiting time at intersections.

III. PROPOSED APPROACH

The predicted travel time of the vehicle along the route is calculated as the sum of the predicted travel time along links of the road network and the predicted waiting time at the regulated intersection.

To estimate the travel time along the road links, it is proposed to use a neural network model. To estimate the waiting time, a traffic light phase prediction is performed at the vehicle arrival time at the intersection. The waiting time is calculated based on the vehicle arrival time and the phase duration time. Let us describe each part of the proposed solution in more detail.

A. Travel Time Prediction

To estimate travel time $t(c)$ of the vehicle c along the road network link, it is proposed to use a model based on an artificial neural network (ANN). The following parameters are applied as input features:

- the distance from the current position of the vehicle to the intersection,
- the vehicle speed,
- the vehicle acceleration,
- the maximum allowed speed on the road network link,
- the number of vehicles in front of the selected one on the network link,

- the turn type at the intersection,
- the position and the speed of the first vehicle on the adjacent outgoing road link.

The architecture of the used ANN model is shown in Fig. 1. The proposed architecture consists of seven linear (fully connected) layers. The output value of the ANN model is the travel time along the road link.

In this paper, we evaluate two approaches to calculate the travel time along the vehicle route. In the first baseline approach, we train two ANNs with the same architecture: for road links adjacent to regulated and unregulated intersections. The ANN model trained on traffic data from regulated road links predicts travel time that implicitly contains the average waiting time at the intersection (according to the training dataset).

In the second approach, we propose to explicitly calculate the waiting time at the intersection according to the predicted phase of the traffic light and the phase duration. To do this, we train the ANN model on traffic data only for vehicles that are moving on the road link when the green light for that link is on. In other words, data from vehicles with waiting times at the intersections is not included in the training dataset.

To estimate the waiting time at the intersection, it is necessary to predict the phase of the traffic light at the time moment when the vehicle arrives at the intersection. In this paper, it is assumed that the traffic light is controlled by an adaptive traffic light control algorithm. The algorithm is described in the next subsection.

B. Adaptive Traffic Light Control

As adaptive control of traffic light phases, a method based on maximizing the predicted weighted flow of vehicles crossing an intersection at a selected traffic light phase is used [4, 5]. In a pseudocode form, this algorithm can be described as follows:

Algorithm: Adaptive traffic light control method

Input data: τ_{min} , t_p , p_{cur} , P

Output data: p_{new}

if $t_p < \tau_{min}$ **then**

$p_{new} = p_{cur}$

$t_p = t_p + 1$

else

$p_{new} = \text{argmax}(\{PredFlow(p) \text{ for } p \text{ in } P\})$

$t_p = 0$

end if

In the algorithm, p_{new} is a new traffic light phase selected from a set of phases P of the considered traffic light, τ_{min} is a phase switching interval, t_p is a duration of the current active phase p_{cur} of the considered traffic light.

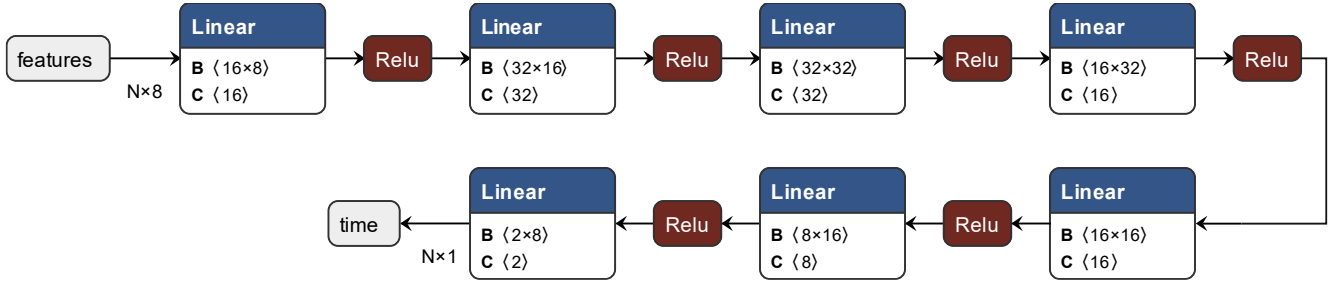


Fig. 1. The architecture of the ANN model.

The main part of the algorithm is the traffic flow estimation, which is performed in the method $PredFlow(p)$ as follows:

$$PredFlow(p) = \sum_l \sum_c I(t(c) < \tau_{min}), \quad (1)$$

where l is a set of incoming road lanes of the considered intersection for which the movement is allowed when the phase $p \in P$ is on, $I(val)$ is an indicator function.

The method $PredFlow(p)$ estimates the number of vehicles passing through the intersection during the specified time interval τ_{min} . The travel time $t(c)$ of the vehicle c is predicted using the model described in Section III.A.

C. Waiting Time Prediction

The estimate of the vehicle's waiting time at a regulated intersection is calculated from the duration of the predicted traffic light phase at the moment the vehicle arrives at the intersection.

The predicted phase is selected based on the following steps:

- For each time moment of phase selection, an estimate of the number of vehicles passed the intersection in the next time interval for each phase is formed. The estimate is based on the predicted travel time from the vehicle position to the nearest intersection obtained by the model from section III.A.
- The traffic light phase for which the estimate of the number of vehicles is maximum is selected.

The waiting time at the intersection for each vehicle is equal to zero if the movement on the corresponding road link is allowed for the predicted traffic light phase. Otherwise, the waiting time is equal to the remaining duration of the traffic light phase.

In the next section, we provide an experimental study of the proposed approach.

IV. EXPERIMENTAL STUDIES

To carry out an experimental study of the proposed approach, the SUMO modeling package was used [22]. In the study, two simulation scenarios are evaluated:

- "Cologne-8" is a small urban area with 8 traffic lights.

- "Cologne-316" is a small traffic scenario that contains 2928 intersections of various topologies, including 316 regulated intersections.

The traffic simulation was carried out from 7:00 to 9:00 during the morning rush hours.

Three following algorithms were compared:

- A deterministic model [20] that predicts the travel time based on the analytical patterns of vehicle movement.
- ANNs that represent two models trained on traffic data for road links adjacent to regulated and unregulated intersections.
- ANN taking into account the waiting time, which is the model presented in the work:

The models were compared according to the criterion of mean absolute error (in seconds) that can be formulated as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |T^{travel} - \tilde{T}^{travel}|, \quad (2)$$

where T^{travel} is the true travel time along the road links, $\tilde{T}^{travel}$ is the predicted travel time, n is the total number of test samples.

The comparison results of the evaluated models are presented in Table I.

TABLE I. COMPARISON OF MODELS BY THE MEAN ABSOLUTE ERROR CRITERION (IN SECONDS)

Model	Scenarios	
	Cologne-8	Cologne-316
Deterministic model	13.88	65.73
ANN	7.92	26.25
ANN + waiting time	7.62	23.92

As can be seen from the table, the proposed approach with the explicit waiting time calculation shows the best result according to the MAE criterion for both scenarios.

At the next step, the total accumulated travel time for all road links of the vehicle's route is compared. Fig. 2 shows the dependence of the mean absolute error of the accumulated predicted travel time on the accumulated true travel time.

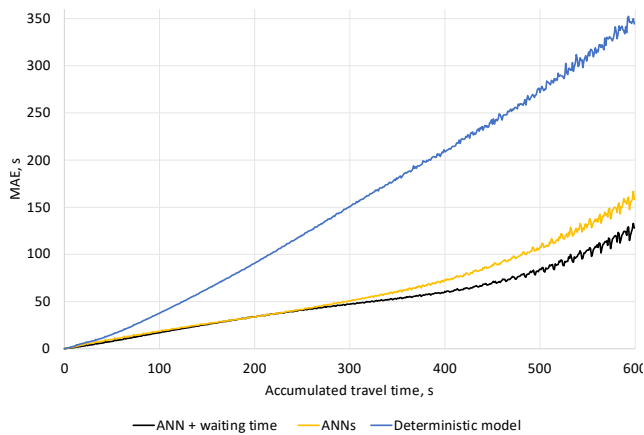


Fig. 2. Comparison of the predicted and observed accumulated travel time.

Experimental results allow us to conclude that the proposed approach has a higher prediction accuracy than the baseline models.

V. CONCLUSION

In this paper, we consider the problem of predicting the travel time of movement of individual connected vehicles in a scenario with adaptive control of traffic lights. It is proposed to use the model based on a neural network to estimate the travel time along the road links of the transportation network, predict the traffic light phase using the adaptive control method, and estimate the waiting time at the intersection based on the predicted arrival time and the duration of the traffic light phase. According to the experimental results, the considered approach makes it possible to reduce the prediction error and to increase the accuracy compared to the baseline methods.

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